



Optimization Techniques for Multi Cooperative Systems MCTR 1021
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Flocking Control of Multi-Robot Systems

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This is to certify that:

- (i) The report comprises only our original work toward the course project.
- (ii) Due acknowledgment has been made in the text to all other material used.

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Abstract

The abstract of the document is added here...
It is usually written after finishing writing all the other chapters.

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Chapter 1

Introduction

Multi-robot systems (MRS) employ the idea that there is strength in numbers. The advantages of using multiple robots to perform a certain task become especially noticeable when the task at hand requires multiple simultaneous operations. In this task, the restrictions of a single robot system can be overcome by MRS. In several applications such as exploration, a large number of simple and cheap robots could have better performance than an individual, highly specialized, and costly robot.

Another advantage of MRS is redundancy as tasks can be reallocated to different robots in case of failure of one or more of the robots. This can increase the reliability of the system.

Chapter 2

Literature Review

In [1], the authors provide a distributed flocking control algorithm achieving the three Reynolds's rules; cohesion, alignment, separation. The contribution is minimizing the conflict between cohesion and separation by relaxing the position synchronization constraint. The authors formulate their problem by considering a network of n rigid bodies in 2D Space. They set 2 objective functions: the cohesion and alignment objectives can be considered as pose synchronization and rigid body i and rigid body j are said to collide with each other if the difference between their poses is less than their diameter.

A relaxation parameter δ_i is introduced in the constraints since they aim to achieve position synchronization as much as possible. In simulation, they plot the 3D position trajectories of the rigid body. In results, the proposed flocking algorithm successfully achieves the desired cohesion, alignment, and separation behaviors for the rigid body network.

In [2], a private flocking optimization problem is tackled. The goal is to generate a private flocking that hides the identity of the leader so that an adversary could not identify the leader. The contributions are formulating the problem in homogeneous mobile robot teams with private leaders, developing an adversarial co-optimization method and providing a study of the relationship between flocking and privacy performance.

Two objective functions are set to this problem which are efficient three-dimensional flocking of a team of mobile robots and privacy of this leadership. In the implementation, components of Reynolds's flocking are adopted. Evolutionary approach based on classical genetic algorithms is adopted. Candidate metrics such as velocity correlation accounts for the direction alignment and the magnitude similarity is used. Overall, the findings show that, despite the huge parameter space and growing severity of the adversarial discriminator, the GA is capable of finding extremely effective flocking solutions.

In most flocking protocols each agent is considered to be only collecting information from its neighbors at a certain time, and afterward, it takes motion decisions neglecting the possi-

bility of using predictive intelligence. To make use of individual predictive intelligence among collective behaviors, an MPC has been proposed [3]. The work was evaluated in a simulation where the goal is to minimize the velocity and lattice disagreement. The results suggest that the MPC method was found to have advantages in the convergence efficacy of flocking.

Cooperative control between mobile sensors is essential. Therefore, Flocking control is essential to resolve the problem of self collision. A non-differentiable optimization technique based on the gravitational search algorithm (GSA) was developed in [4]. To evaluate the performance of the proposed method a single target moving with a circular wave, and the multiple targets moving with a semi-circular and semi-sine wave trajectory were simulated. It was found that the center of flocks precisely tracks the moving targets.

The Ant Colony Optimisation (ACO) is one of the mathematically mature optimisation algorithms based on the social behaviour of animal colonies. An ACO Based Hybrid Algorithms was reviewed to be used for Path Planning of Multi-Mobile Robotics in [5]. It was found that The original ACO performs well in complex combinatorial optimisation problems but it has convergence speed and local optima problems. Based on the presented review and survey they were solved by embedding Genetic, A*, Cellular automata, Improved Potential Field and Fuzzy algorithms.

In [6] the authors tackle the problem of flocking control with obstacle avoidance. The mentioned problem is considered a many-objective optimization problem for centralized algorithms. In this work the authors render the many-objective problem into a distributed multi-objective optimization algorithm. There are four objectives for this flocking problem which are: if obstacles are detected pass through them, keep a steady distance with each neighbour, keep a safe distance away from obstacles and avoid collision with neighbours.

The authors divide the four objective into soft constraints (1 and 2) and hard constraints (3 and 4). A modified pigeon-inspired optimization technique is proposed. The problem formulation in this paper starts with a kinematic model for a UAV and sets a number of constraints for horizontal speed and altitude rate. Comparative studies between the Modified MPIO, classic MPIO and NSGA-II were conducted which showed that the modified MPIO is superior.

In [7] the authors develop a flocking technique based on the behavior that pigeons will switch between hierarchical and egalitarian interaction modes at different flight phases. The proposed controller is capable of coordinating the flight of a swarm of UAVs through obstacle environments with few informed individuals. The paper uses a dynamic model with control input that consists of a number of parameters similar to previously discussed work that handle cohesion, alignment and obstacle avoidance. Simulations were conducted and the swarm with very few informed individuals was able to maintain a flock formation and navigate through obstacles. The proposed flocking controller is superior in state convergence time, time reaching target position and trajectory smoothness.

In [8], the authors presented a new flocking model for autonomous flying agents. The four measures of coherence, collision avoidance, obstacle avoidance, and the flocking speed are used to evaluate the performance. They accepted local coherence and correlated sub-flock clusters without global coherence as a good solution. The second measure calculates the number of collisions between the agents by using a characteristic distance above which a collision occurs. The same criteria applies for the obstacle avoidance. The flocking speed is determined by taking the average of the velocities of all agents.

The fitness function used here was resulted from multiplying all fitness functions computed for all the parameters discussed before. The fitness functions used were one of three types, a monotonically growing function, a single maximum at zero and a smooth decay around it or a sharp peak. The decision variables in this optimization problem are 11 variables coming from their new model to achieve the required flocking.

A hybrid flocking controller is presented in [9]. The hybrid flocking control of UAVs is achieved by using three types of guidance methods: vector field, Chucker-Smale model, and potential field. The usage of several methods in hybrid flocking may lead to conflict command generation which degrades the performance of the mission. This issue is addressed by presenting the adaptive Chucker-Smale model. The optimal weightings between guidance methods is determined using social learning particle swarm optimization (SL-PSO). The research uses a dynamic model for his UAV with control inputs that consist of desired speed, heading angle and target point. After which the different guidance methods are formalized. The weighting is done using PSO with 4 different Fitness functions that control cohesion, alignment, velocity matching and obstacle avoidance. The final fitness function is the result of multiplication of the previous fitness functions and tries to maximize it. Simulations were conducted and showed that the proposed adaptive augmented CS model improves the mission performance in comparison to the conventional CD model.

In [10], a new genetic flocking optimizing framework is designed. This framework is able to achieve both stability and adaptivity of robotic swarms. In this work the robot is separated into three layers: sensing, decision and action. These layers support the basic navigation functions. The environment on the other hand is divided into two layers: the evaluation and evolutionary layers where the evaluation layer provides fitness values for the evolutionary one. The researchers propose a velocity update function which take into account Reynolds' definition of flocking maintaining cohesion, separation and alignment. This formulation is used to create a single objective optimization problem through calculating the variation range of an-isotropic index. This allows using GA to optimize this problem. The resulting model is then simulated and compared with basic rule-based model. The new model performs better than the basic rule-based model in uniformity and stability.

Chapter 3

Problem Formulation

In this chapter, we will present the decision variables that we are basing our problem on, the objective function which is the goal that we aspire to reach, and the constraints that our system is subject to them. The problem consists of n nonholonomic differential mobile robots. The robots operate in a continuous map that has a length L and a width W . The map contains a number O of obstacles, each i_{th} obstacle has radius $r_{o \rightarrow i}$. The robots start around point S and their goal is to reach point G while flocking. According to Reynolds, there are three main components of flocking [1]:

- Cohesion
- Separation
- Alignment

To be more specific robots should maintain a close distance between each other (Cohesion), robots also match their velocities (Alignment) and avoid collision among themselves (Separation).

Here we define several variables which are crucial to our formulation:

- r_{bi} is the radius of the i_{th} robot,
- r_{ij} is the distance between the centres of robot i and robot j where $i, j \leq n$,
- $d_{o \rightarrow j}$ is the distance between the centres of robot j and its closest obstacle,
- $\sigma(V)$ is the standard deviation of the n robots' linear velocities.

3.1 Decision Variables

The goal of this optimization problem is to find n paths K for the n robots to follow from the start point S to the end goal G . The paths will consist of a number of points N_p for each

robot. Each point is a tuple containing the x position and the y position of the i_{th} robot $(x, y)_i$.

3.2 Objective Functions

To achieve the desired behavior this section presents the There are four objective functions in our problem. The mobile robots try to reach our final destination represented by point G whilst flocking.

The first one is to keep the distance between each two robots around a certain value r_f . This addresses the cohesion and separation part of the flocking problem.

$$\operatorname{argmin}_K \sum_{i=1}^n |r_{ij} - r_f| \quad (3.1)$$

The second objective function deals with matching the velocities of the robots. This is related to alignment. This is done by minimizing the standard deviation of the robots' velocities.

$$\operatorname{argmin}_K \sigma(V) \quad (3.2)$$

The third objective function is to avoid obstacles. The obstacles are defined as circles regardless of their shapes with a radius r_o .

$$\operatorname{argmax}_K \sum_{i=1}^n |d_{o \rightarrow j} - (r_{oi} + r_{bi}) - c| \quad (3.3)$$

The fourth objective function is minimizing the change in the steering angles of the mobile robots to avoid sudden changes in trajectory. θ_i defines the change in the angle of the i_{th} robot.

$$\operatorname{argmin}_K \sum_{i=1}^n \Delta\theta_i \quad (3.4)$$

The cost function is defined as summing all previous elements elements multiplied by four different weights.

$$cost = w_1 * \sum_{i=1}^n |r_{ij} - r_f| + w_2 * \sigma(V) + w_3 * \sum_{i=1}^n |d_{o \rightarrow j} - (r_o + r_b) - c| + w_4 * \sum_{i=1}^n \Delta\theta_i \quad (3.5)$$

3.3 Constraints

The first constrain is to stay inside the map.

$$0 < P_{ix} < L \quad \forall i \quad (3.6)$$

$$0 < P_{iy} < W \quad \forall i \quad (3.7)$$

The second constraint is that the velocity of each robot must not exceed the maximum velocity V_{max}

$$V_i < V_{max} \quad \forall i \quad (3.8)$$

The third constraint is the steering angle. The steering angle is not allowed to exceed a maximum limit δ_f

$$\theta_i < \delta_f \quad \forall i \quad (3.9)$$

3.4 Mobile Robot Kinematics

The differential robot that will be used consists of two back wheels that are responsible for controlling the robot motion and one passive front wheel for balancing. Since the robot can move freely in a 2D-plane, it has 3 DOF which requires 3 independent parameters to describe its location. This kinematic model will consider three variables and their rate of change:

- θ : The rotation angle of the differential robot relative to the $+ve$ x-axis
- (x, y) : The location of a fixed point on the differential robot relative to a fixed frame on the map. In this model, this point is the middle point between the two back wheels.

Since the robot performs general plane motion, its velocity can be described by two velocity vectors, the velocity vector of the chosen fixed point v_T in addition to the whole robot angular velocity vector ω (figure 3.1a).

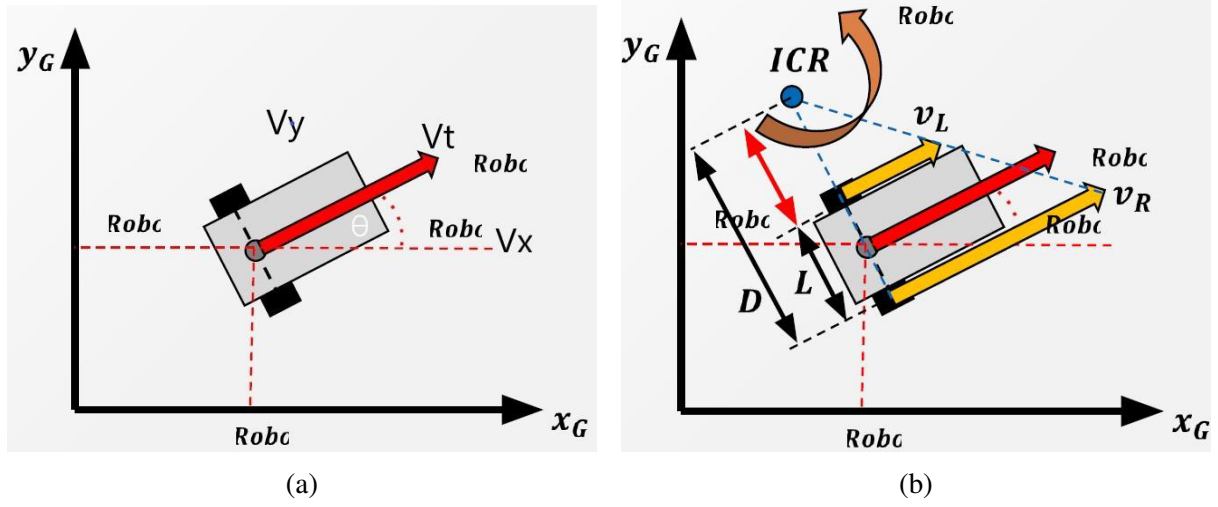


Figure 3.1: Visual kinematic analysis of the differential mobile robot

The velocity of the fixed point can be obtained from the velocity of the right and left wheels (figure 3.1b).

$$v_T = \frac{v_R + v_L}{2} \quad (3.10)$$

$$v_x = v_T \cos \theta, \quad v_y = v_T \sin \theta$$

The velocity of the right and left wheels can be obtained from the angular velocity of each wheel (figure 3.1b).

$$v_R = \omega_R \times r \quad (3.11)$$

$$v_L = \omega_L \times r$$

Where ω_R, ω_L, r are the angular velocity of the right wheel, the angular velocity of the left wheel and the radius of the wheel, respectively. By substituting from 3.11 in 3.10:

$$v_T = \frac{\omega_R \times r + \omega_L \times r}{2} = \frac{r}{2}(\omega_R + \omega_L) \quad (3.12)$$

$$v_x = \frac{r \cos \theta}{2}(\omega_R + \omega_L) \quad (3.13)$$

$$v_y = \frac{r \sin \theta}{2}(\omega_R + \omega_L) \quad (3.14)$$

Equations (3.13) and (3.14) describe the x and y components of the velocity vector for the fixed point. The angular velocity of the whole robot ω can be obtained from the following

equations that are obtained from figure 3.1b.

$$v_R = \omega \times D \quad , \quad v_L = \omega \times (D - L)$$

Solving the two equations for ω :

$$\omega = \frac{v_R - v_L}{L} = \frac{r}{L}(\omega_R - \omega_L) = \dot{\theta} \quad (3.15)$$

Equations (3.13), (3.14) and (3.15) are used to control the motion of the robot.

Chapter 4

Results

Chapter 5

Conclusion

Chapter 6

Future Work

Appendix

Bibliography

- [1] Tatsuya Ibuki, Sean Wilson, Junya Yamauchi, Masayuki Fujita, and Magnus Egerstedt. Optimization-based distributed flocking control for multiple rigid bodies. *IEEE Robotics and Automation Letters*, 5(2):1891–1898, 2020.
- [2] Hehui Zheng, Jacopo Panerati, Giovanni Beltrame, and Amanda Prorok. An adversarial approach to private flocking in mobile robot teams. *IEEE Robotics and Automation Letters*, 5(2):1009–1016, 2020.
- [3] Lifeng Zhou and Shaoyuan Li. Distributed model predictive control for multi-agent flocking via neighbor screening optimization. *International Journal of Robust and Nonlinear Control*, 27(9):1690–1705, 2017.
- [4] E Khodayari, V Sattari-Naeini, and M Mirhosseini. A gravitational search algorithm-based single-center of mass flocking control for tracking single and multiple dynamic targets for parabolic trajectories in mobile sensor networks. *Journal of AI and Data Mining*, 6(1):207–217, 2018.
- [5] Ibrahim Hamad and Mohammad Hasan. A review: On using aco based hybrid algorithms for path planning of multi-mobile robotics. 2020.
- [6] Huaxin Qiu and Haibin Duan. A multi-objective pigeon-inspired optimization approach to uav distributed flocking among obstacles. *Information Sciences*, 509:515–529, 2020.
- [7] Huaxin Qiu and Haibin Duan. Pigeon interaction mode switch-based uav distributed flocking control under obstacle environments. *ISA transactions*, 71:93–102, 2017.
- [8] Gábor Vásárhelyi, Csaba Virágh, Gergő Somorjai, Tamás Nepusz, Agoston E Eiben, and Tamás Vicsek. Optimized flocking of autonomous drones in confined environments. *Science Robotics*, 3(20), 2018.
- [9] Seunghan Lim, Yeongho Song, Joonwon Choi, Hyunsam Myung, Heungsik Lim, and Hyondong Oh. Decentralized hybrid flocking guidance for a swarm of small uavs. In *2019 Workshop on Research, Education and Development of Unmanned Aerial Systems (RED UAS)*, pages 287–296. IEEE, 2019.

- [10] Li Ma, Weidong Bao, Xiaomin Zhu, Meng Wu, Yuan Wang, Yunxiang Ling, and Wen Zhou. G-flocking: Flocking model optimization based on genetic framework. *ArXiv*, abs/1907.11852, 2019.